K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

Big Data Analysis with Scala and spark Lab

(ENSP359)

Lab File

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Submitted to

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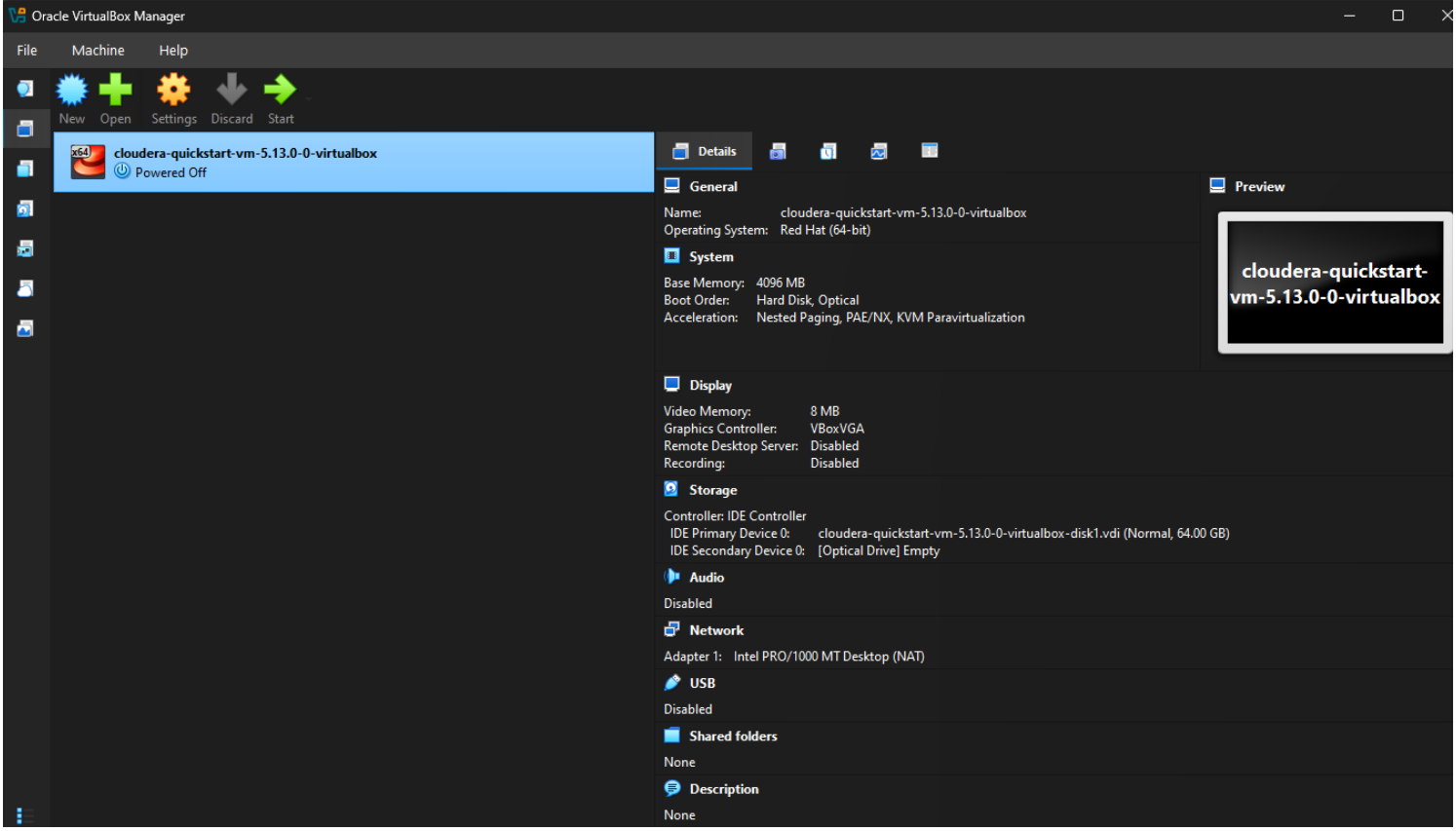
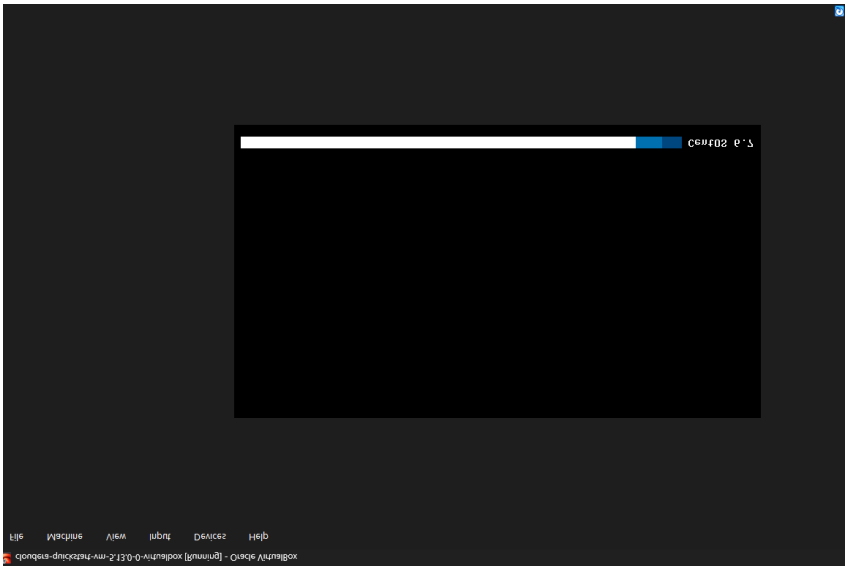
Task 1

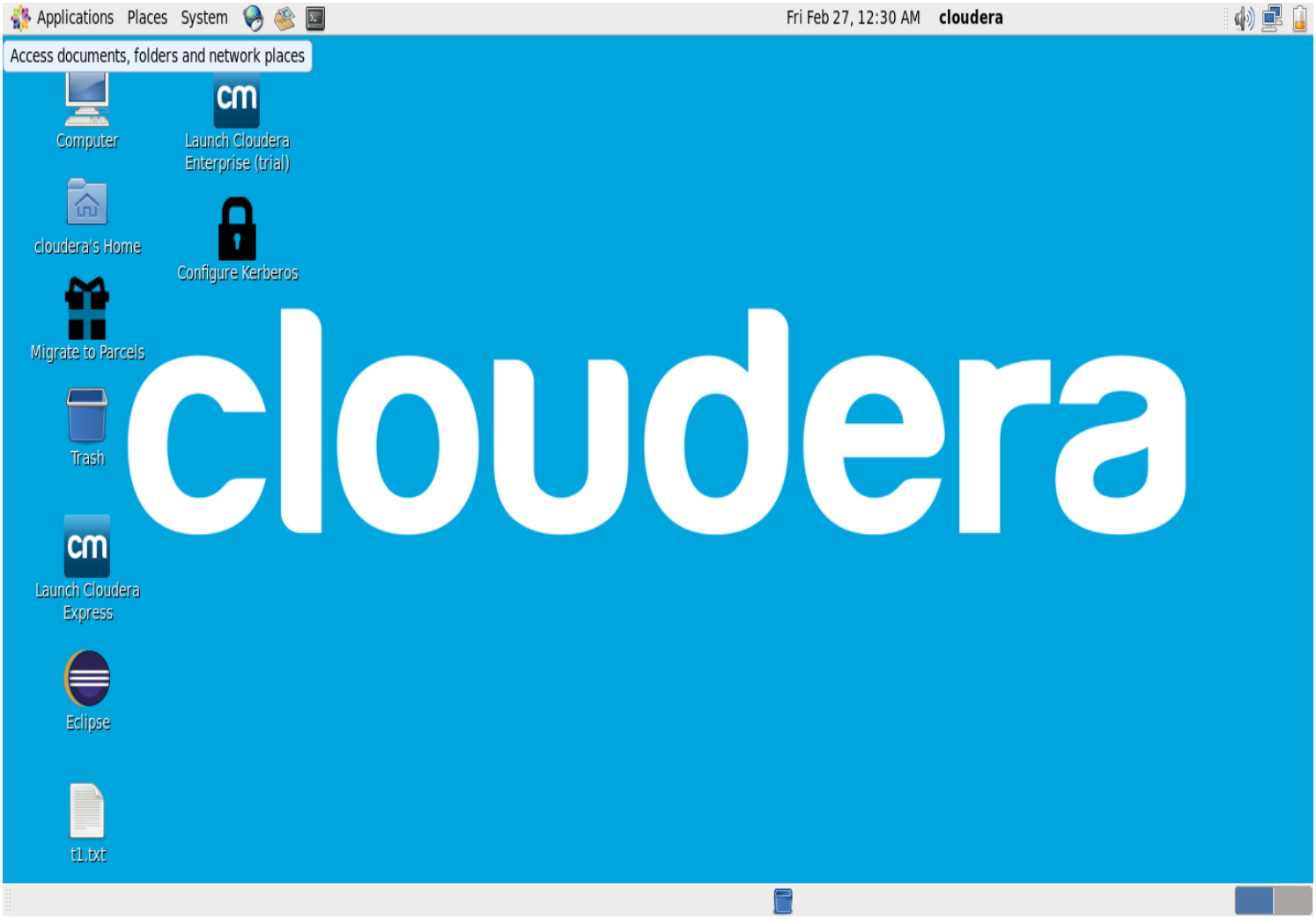


VirtualBox Main Screen

This screenshot shows the Cloudera QuickStart VM (v5.13.0) successfully imported in Oracle VirtualBox with 4096 MB RAM, 64 GB storage, and Red Hat64-bit operating system.

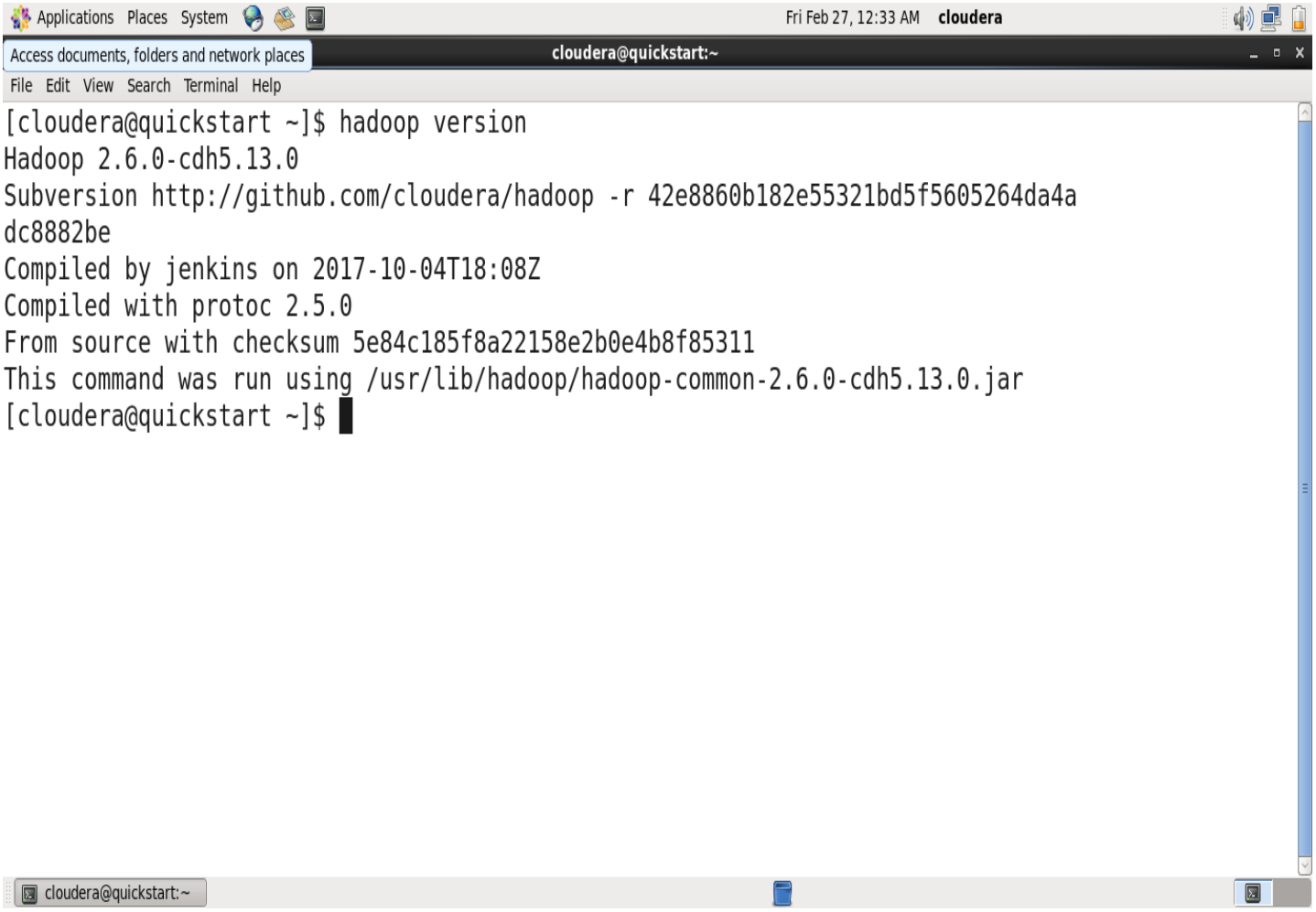


Starting Process



Cloudera Desktop

Cloudera QuickStart VM successfully booted and logged in as 'cloudera' user. The desktop shows pre-installed tools like Cloudera Manager, Eclipse, and other Big Data utilities.



Hadoop version

The 'hadoop version' command confirms that Hadoop 2.6.0-cdh5.13.0 is successfully installed and configured in the Cloudera QuickStart environment.

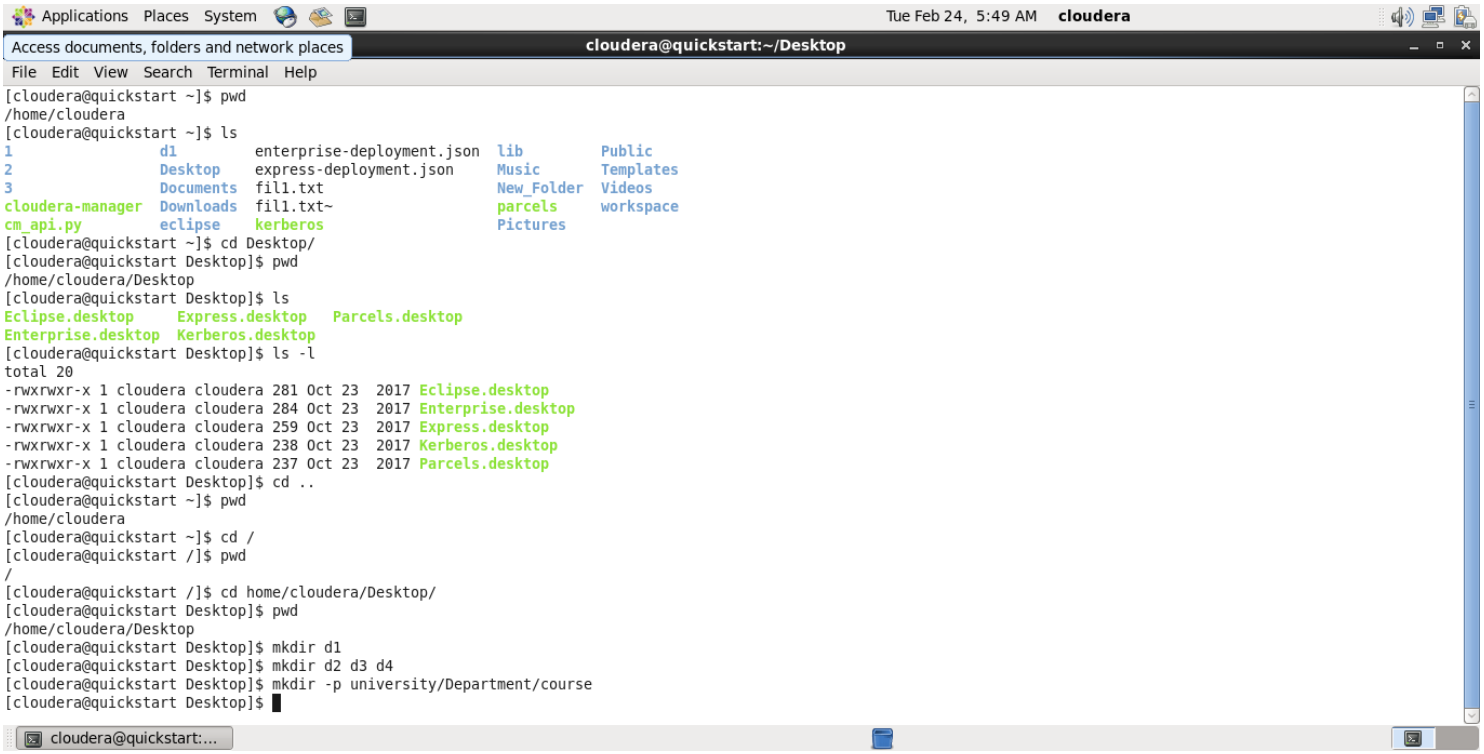
```
[cloudera@quickstart ~]$ spark-shell
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel).
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/zookeeper/lib/slf4j-log4j12-1.7.5.jar!/org/slf4j/impl/StaticLoggerBin
SLF4J: Found binding in [jar:file:/usr/lib/flume-ng/lib/slf4j-log4j12-1.7.5.jar!/org/slf4j/impl/StaticLoggerBind
SLF4J: Found binding in [jar:file:/usr/lib/parquet/lib/slf4j-log4j12-1.7.5.jar!/org/slf4j/impl/StaticLoggerBinde
SLF4J: Found binding in [jar:file:/usr/lib/avro/avro-tools-1.7.6-cdh5.13.0.jar!/org/slf4j/impl/StaticLoggerBinde
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /___/ .__/\_,_/_/ /_/\_\   version 1.6.0
      /_/

Using Scala version 2.10.5 (Java HotSpot(TM) 64-Bit Server VM, Java 1.7.0_67)
Type in expressions to have them evaluated.
Type :help for more information.
26/02/24 08:03:58 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using bu
Spark context available as sc (master = local[*], app id = local-1771949040570).
26/02/24 08:04:03 WARN shortcircuit.DomainSocketFactory: The short-circuit local reads feature cannot be used be
SQL context available as sqlContext.
```

spark-shell

Apache Spark Shell (version 1.6.0) successfully launched using Scala 2.10.5. Spark context (sc) and SQL context (sqlContext) are automatically initialized and ready for use.

Task 2



Command – pwd

The 'pwd' command displays the current working directory. Output shows /home/cloudera.

Command – ls

The 'ls' command lists all files and folders in the current directory.

Command – cd

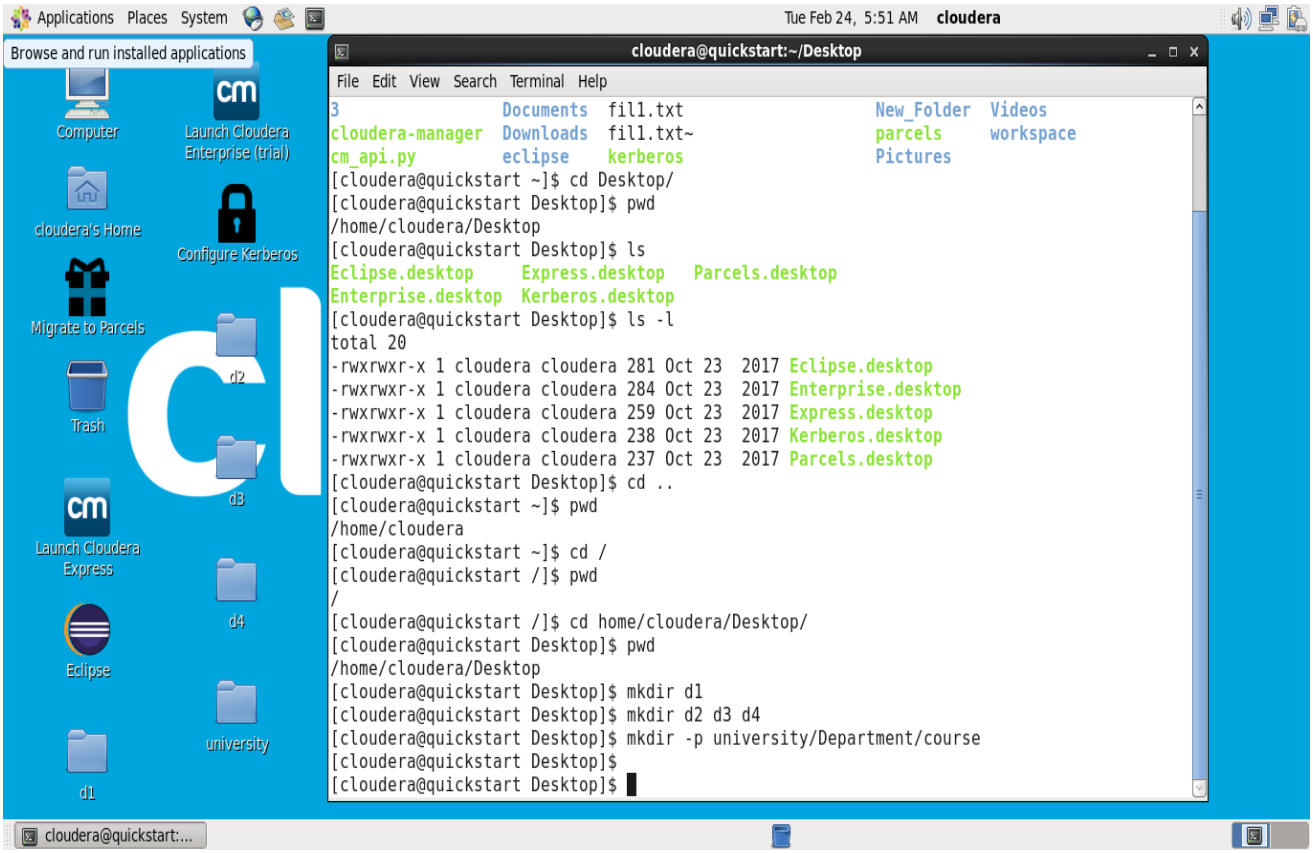
The 'cd Desktop/' command changes the current directory to Desktop. 'cd ..' goes back one level and 'cd /' goes to root directory."

Command – ls -l

The 'ls -l' command shows detailed list of files with permissions, owner, size and date.

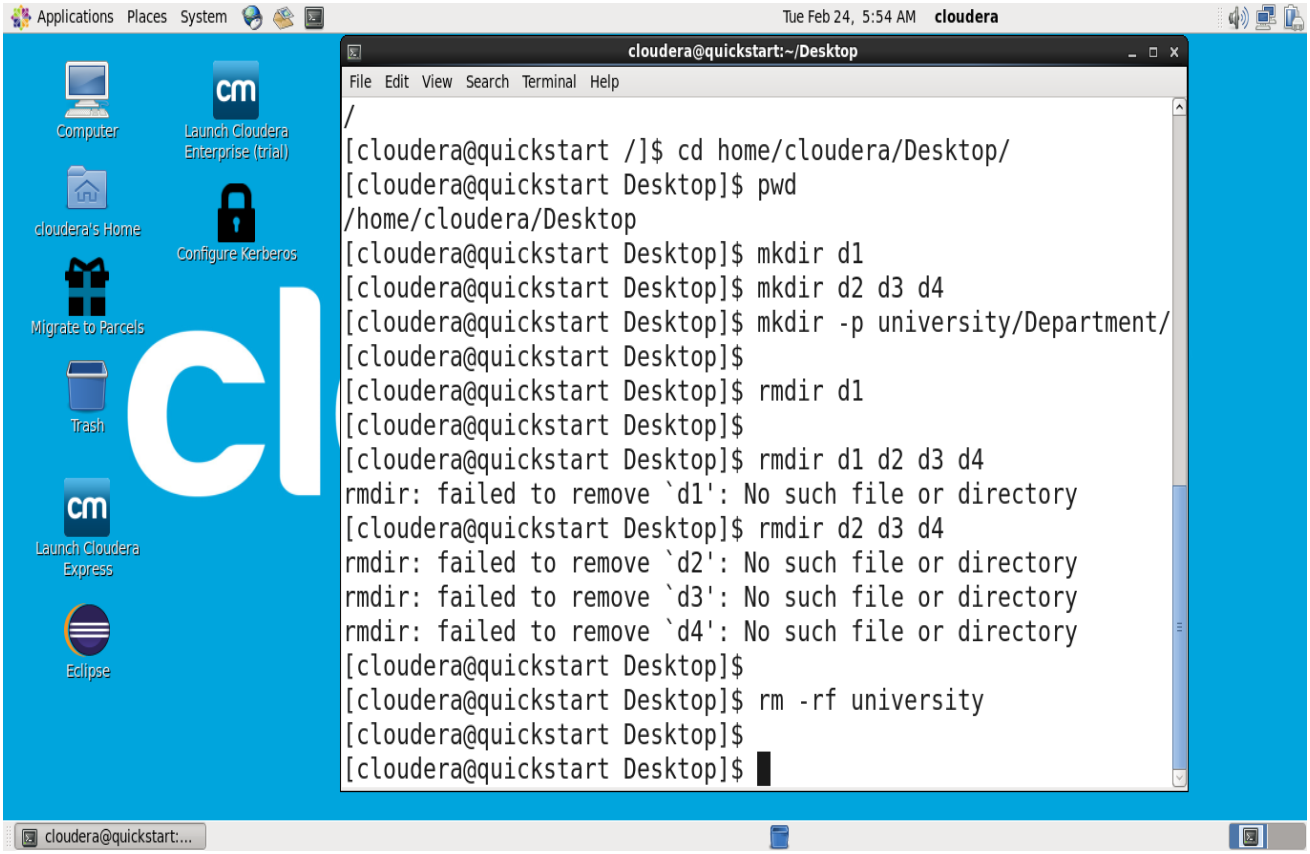
Command – mkdir

The 'mkdir' command creates new directories. 'mkdir -p university/Department/course' creates nested directories at once.



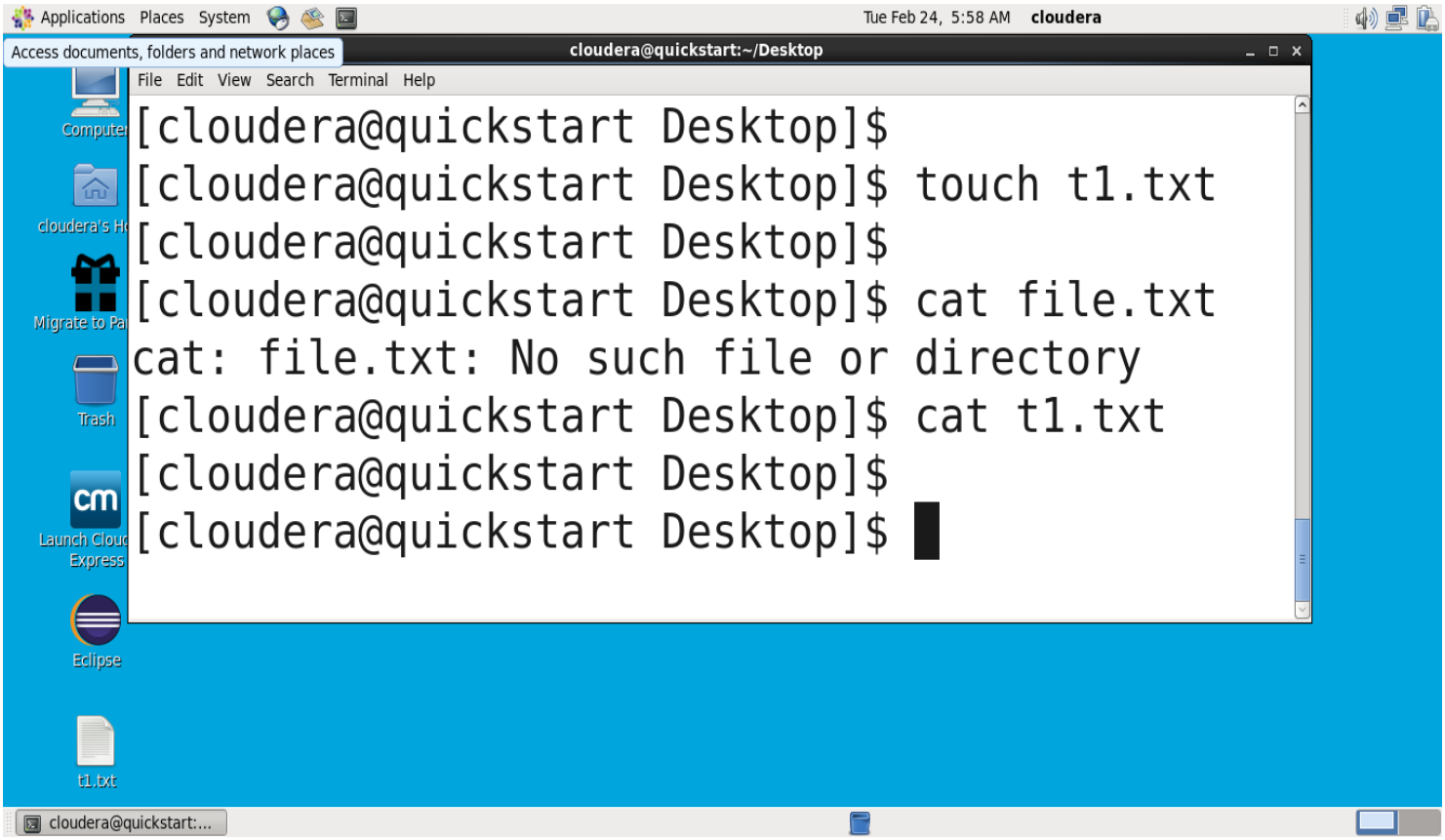
This screenshot shows the result of mkdir commands.

Directories d1, d2, d3, d4 and nested directory 'university/Department/course' are successfully created, visible on the Desktop.



The 'rmdir' command removes empty directories.

The 'rm -rf university' command forcefully removes the directory along with all its contents recursively



The 'touch t1.txt' command creates an empty file. The 'cat t1.txt' command displays file contents (empty as no data written yet). Desktop also shows t1.txt file created successfully

```
[cloudera@quickstart ~]$ pwd
/home/cloudera
[cloudera@quickstart ~]$ cd Desktop/
[cloudera@quickstart Desktop]$
[cloudera@quickstart Desktop]$ pwd
/home/cloudera/Desktop
[cloudera@quickstart Desktop]$ hdfs dfs -ls /
Found 7 items
drwxr-xr-x   - cloudera supergroup          0 2026-02-19 20:46 /New_folder
drwxrwxrwx   - hdfs     supergroup          0 2017-10-23 09:15 /benchmarks
drwxr-xr-x   - hbase    supergroup          0 2026-02-24 07:33 /hbase
drwxr-xr-x   - solr     solr                0 2017-10-23 09:18 /solr
drwxrwxrwt   - hdfs     supergroup          0 2026-01-22 21:17 /tmp
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /user
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /var
[cloudera@quickstart Desktop]$ cd ..
[cloudera@quickstart ~]$ pwd
/home/cloudera
[cloudera@quickstart ~]$ hdfs dfs -ls /
Found 7 items
drwxr-xr-x   - cloudera supergroup          0 2026-02-19 20:46 /New_folder
drwxrwxrwx   - hdfs     supergroup          0 2017-10-23 09:15 /benchmarks
drwxr-xr-x   - hbase    supergroup          0 2026-02-24 07:33 /hbase
drwxr-xr-x   - solr     solr                0 2017-10-23 09:18 /solr
drwxrwxrwt   - hdfs     supergroup          0 2026-01-22 21:17 /tmp
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /user
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /var
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$ hdfs dfs -mkdir /data
[cloudera@quickstart ~]$ cd ..
```

The 'hdfs dfs -ls /' command lists all directories in HDFS (Hadoop Distributed File System). It shows 7 items including /benchmarks, /hbase, /tmp, /user etc. The 'hdfs dfs -mkdir /data' command creates a new directory '/data' in HDFS.

```
[cloudera@quickstart ~]$ pwd
/home/cloudera
[cloudera@quickstart ~]$ hdfs dfs -mkdir /new_folder
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$ hdfs dfs -ls /
Found 9 items
drwxr-xr-x   - cloudera supergroup          0 2026-02-19 20:46 /New_folder
drwxrwxrwx   - hdfs     supergroup          0 2017-10-23 09:15 /benchmarks
drwxr-xr-x   - cloudera supergroup          0 2026-02-24 07:48 /data
drwxr-xr-x   - hbase    supergroup          0 2026-02-24 07:33 /hbase
drwxr-xr-x   - cloudera supergroup          0 2026-02-24 08:00 /new_folder
drwxr-xr-x   - solr     solr                0 2017-10-23 09:18 /solr
drwxrwxrwt   - hdfs     supergroup          0 2026-01-22 21:17 /tmp
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /user
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /var
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$ hdfs dfs -ls /new_folder
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$ hdfs dfs -ls /
Found 9 items
drwxr-xr-x   - cloudera supergroup          0 2026-02-19 20:46 /New_folder
drwxrwxrwx   - hdfs     supergroup          0 2017-10-23 09:15 /benchmarks
drwxr-xr-x   - cloudera supergroup          0 2026-02-24 07:48 /data
drwxr-xr-x   - hbase    supergroup          0 2026-02-24 07:33 /hbase
drwxr-xr-x   - cloudera supergroup          0 2026-02-24 08:00 /new_folder
drwxr-xr-x   - solr     solr                0 2017-10-23 09:18 /solr
drwxrwxrwt   - hdfs     supergroup          0 2026-01-22 21:17 /tmp
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /user
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 09:17 /var
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$ hdfs dfs -rm -r /new_folder
Deleted /new_folder
[cloudera@quickstart ~]$
```

The 'hdfs dfs -mkdir /new_folder' creates a new directory in HDFS. 'hdfs dfs -ls /' lists all HDFS directories showing 9 items including /data and /new_folder. 'hdfs dfs -rm -r /new_folder' deletes the directory from HDFS.

```
This command was run using /usr/lib/hadoop/hadoop-common-2.6.0-cdh5.13.0.jar
[cloudera@quickstart ~]$ echo "Hello Big Data">
t1.txt
[cloudera@quickstart ~]$ cp t1.txt t1_backup.tx
t
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$ mv t1_backup.txt t1_re
named.txt
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$ ls -l t1.txt
-rw-rw-r-- 1 cloudera cloudera 15 Feb 27 01:38
t1.txt
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$ grep "Hello" t1.txt
Hello Big Data
[cloudera@quickstart ~]$
[cloudera@quickstart ~]$
```

Command -echo & cp :

The 'echo "Hello Big Data"' command writes text into t1.txt file. The 'cp t1.txt t1_backup.txt' command creates a copy of the file.

Command -Mv:

The 'mv t1_backup.txt t1_renamed.txt' command renames/moves the file from t1_backup.txt to t1_renamed.txt.

Command -ls –l:

The 'ls -l t1.txt' command shows detailed information of the file including permissions (-rw-rw-r--), owner (cloudera), size (15 bytes) and date.

Command -grep :

The 'grep "Hello" t1.txt' command searches for the word 'Hello' in t1.txt and displays the matching line: Hello Big Data.

```
scala> val num1=Vector.range(1,11)
num1: scala.collection.immutable.Vector[Int] = Vector(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)

scala> println(num1.head)
1

scala> for(i <- 1 to 20){println(i)}
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
```

The 'Vector.range(1,11)' creates an immutable Vector with values 1 to 10.

println(num1.head) = prints the first element (1).

The for loop 'for(i <- 1 to 20)' prints numbers 1 to 20.

```
17
18
19
20

scala> num1.foreach(println)
1
2
3
4
5
6
7
8
9
10

scala> def add(a:Int,b:Int):Int={a+b}
add: (a: Int, b: Int)Int

scala> add(49,51)
res3: Int = 100

scala> def sub(a:Int,b:Int):Int={a-b}
sub: (a: Int, b: Int)Int
```

num1.foreach(println)' prints each element of the Vector one by one

.def add(a:Int, b:Int):Int={a+b}' defines a function to add two numbers, add(49,51) returns 100.

def sub' defines a subtraction function

```
Applications  Places  System                    Fri Feb 27, 2:05 AM   cloudera
Browse and run installed applications          cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
              clear
              ^

scala> val L1=List.range(1,11)
L1: List[Int] = List(1, 2, 3, 4, 5, 6, 7, 8, 9,
 10)

scala> L1.head
res1: Int = 1

scala> l1.tail
<console>:26: error: not found: value l1
              l1.tail
              ^

scala> L1.tail
res3: List[Int] = List(2, 3, 4, 5, 6, 7, 8, 9,
10)

scala> val arr=Array(23,23,13,43,54)
arr: Array[Int] = Array(23, 23, 13, 43, 54)

scala> arr.mkString(", ")
res4: String = 23, 23, 13, 43, 54

scala> arr(0)=99

scala> arr.mkString(", ")
res6: String = 99, 23, 13, 43, 54

cloudera@quickstart:~
```

List

List.range(1,11)' creates a List with values 1 to 10.

L1.head' returns the first element (1).

L1.tail' returns all elements except the first: List(2,3,4,5,6,7,8,9,10).

Array

Array(23,23,13,43,54)' creates a mutable Array.

arr.mkString' displays elements separated by comma.

arr(0)=99' updates the first element from 23 to 99, showing Array mutability.

```
scala> val student =Map("Name"->"Raj","Age"->"18","Course"->"BCA")
student: scala.collection.immutable.Map[String,String] = Map(Name -> Raj, Age -> 18, Course -> BCA)

scala> student("Name")
res7: String = Raj

scala> student("Course")
res8: String = BCA

scala> student.keys
res9: Iterable[String] = Set(Name, Age, Course)

scala> student.values
res10: Iterable[String] = MapLike(Raj, 18, BCA)

scala> student.size
res11: Int = 3

scala> student.contains("Age")
res12: Boolean = true

scala> val st1=student + ("Grade"->"A")
st1: scala.collection.immutable.Map[String,String] = Map(Name -> R
aj, Age -> 18, Course -> BCA, Grade -> A)
```

<u>Map</u>

Map 'student' is created with key-value pairs (Name, Age, Course).

student("Name")' returns 'Raj'. 'student("Course")' returns 'BCA'.

student.keys' shows all keys: Set(Name, Age, Course).

student.values' shows all values: (Raj, 18, BCA).

student.size' returns 3 (total pairs).

student.contains("Age")' returns true.

A new Map 'st1' is created by adding Grade->A to existing student Map